

Assignment /

Assignment #: 11

Topic : Modules & Packages

Level 1: Easy

1. Write a Python program to import the math module and use it to calculate the square root of a given number.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:
Input	16	25
Output	4.0	5.0

2. Create a custom module named calculator with functions for addition and subtraction. Write a program to import this module and use its functions.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:
Input	add(2, 3)	subtract(5, 2)
Output	5	3

3. Create a package named geometry with a module shapes containing functions to calculate the area of a circle and a rectangle. Write a program to use these functions.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:
Input	circle_area(3)	rectangle_area(4, 5)
Output	28.27	20

4. Write a Python program to import only the sin and cos functions from the math module and calculate the sine and cosine of 90 degrees.

Sample test Cases

Test Cases	Test Case 1:
Input	90 (degrees)
Output	sin(90) = 1.0 cos(90) = 0.0

Level 2: Medium

1. Create a package named utilities with submodules string_utils and number_utils. The string_utils module should have a function to count vowels in a string, and number_utils should have a function to check if a number is prime. Write a program to use these functions.

Sample test Cases:

Test Cases	Test Case1:	Test Case 2:	Test Case 3:
Input	hello	7	8
Output	2	True	False

2. Write a Python program using the random module to generate a random password of a given length. The password should include letters, digits, and special characters.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:
Input	length=8	length=12
Output	Generated password: 'aB3\$dF7g	Generated password: 'P@5qR2&uT6zX

3. Develop a Python script using the math module to calculate the square root and the factorial of a given number.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:
Input	number=16	number=5
Output	Square root: 4.0, Factorial: 20922789888000	Square root: 2.23606797749979, Factorial: 120

4. Write a Python program to import the random module and select a random item from a list.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:
Input	[1, 2, 3, 4, 5]	['apple', 'banana', 'cherry']
Output	3 (Random item from the list)	banana (Random item from the list)